PREFACE

In an era of rapidly advancing technology and increasing demands on agricultural productivity, the role of innovation in optimizing farming practices has never been more crucial. Aerial imagery, coupled with ”SOTA” machine learning algorithms, offers a promising avenue for revolutionizing agricultural practices. This project delves into the realm of aerial object detection using Convolutional Neural Networks (CNNs), with a particular focus on identifying human presence for emergency response applications.

In today's interconnected world, where rapid response to emergencies can mean the difference between life and death, the ability to swiftly and accurately detect human presence in remote or inaccessible areas is paramount. By harnessing the power of drone technology and CNN models like YOLOv8, this project endeavors to work on a robust system capable of determining humans from aerial imagery.

This report serves as a comprehensive exploration of our methodology, from dataset collection to model training and deployment. This report provides an overview of the methodology employed, the data analysis techniques used, and presents the research findings obtained through the application of different models. The evaluation metrics employed, such as “Map50 & mAP 95”, are discussed to assess the score and performance of models. We discuss the challenges encountered along the way and present our findings with transparency and rigor. By empowering emergency response teams with timely and accurate information, we aspire to make a tangible difference in saving lives and mitigating crises.

Through this project, we not only showcase the potential of aerial object detection but also underscore the ethical responsibility that comes with wielding such technology. As we navigate the complexities of deploying AI in real-world scenarios, we remain committed to upholding the highest standards of integrity, accountability, and social impact.